

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	D	B	B	B	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,B,D	A,B,D	A,B	A,B,C,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.00	5.00	1.35 to 1.43	0.99	14.72	8.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	C	C	B	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,D	A,B,C,D	A,B,D	A,B,C	A,D	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	4.00	7.00	4.00	3.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	D	B	C	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	B,C,D	B,C	B,C	A,B,D	A,C,D	B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.00	1.00	50.00	0.00	1.00	4.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

 1. **Ans (C)**

 The charge on C_1 is

$$q_1 = C_1 V_1 = 3 \times 10^{-6} \times 4000 = 0.012C$$

 The charge on C_2 is

$$q_2 = C_2 V_2 = 6 \times 10^{-6} \times 3000 = 0.018C$$

The combined capacitance

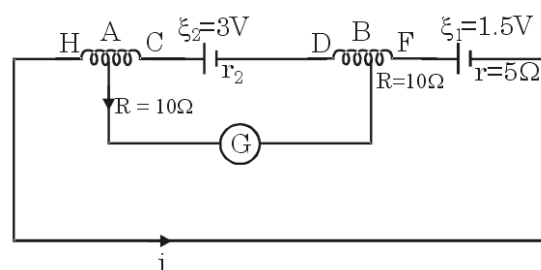
$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{3 \times 6}{3 + 6} \times 10^{-6} = 2 \times 10^{-6} F$$

 Take the lower charge to find the maximum voltage V

$$V = \frac{q_1}{C} = \frac{0.012}{2 \times 10^{-6}} = 6000V = 6.0 kV$$

 2. **Ans (D)**

Let A and B be the midpoints of the coils. As no current flows through the galvanometer, P.D across AB is zero. Applying the loop theorem to the main circuit,



$$\xi_1 + \xi_2 - i(R + R + r_1 + r_2) = 0$$

$$\therefore i = \frac{\xi_1 + \xi_2}{2R + r_1 + r_2} = \frac{1.5 + 3}{2 \times 10 + 5 + r_2} = \frac{4.5}{r_2 + 25} \dots (1)$$

The P.D of the point B with respect to the negative terminal of the first cell is

$$V_B = \xi_1 - \left(\frac{10}{2} + r_1 \right) i = 1.5 - 10i \dots (2)$$

The P.D of the point A with respect to the negative terminal of the first cell is

$$V_A = \xi_1 + \xi_2 - \left(\frac{10}{2} + r_2 + 10 + r_1 \right) i$$

$$= 1.5 + 3 - (5 + r_2 + 10 + 5)i = 4.5 - (20 + r_2)i$$

But $V_B = V_A$ ($\because$ no current flows through the galvanometer)

$$\therefore 1.5 - 10i = 4.5 - (20 + r_2)i$$

$$\therefore i = \frac{3}{10 + r_2} \quad \dots(3)$$

From (1) and (2), $r_2 = 20\Omega$

3. Ans (B)

The induction midway between Helmholtz coils is

$$B = \frac{8N\mu_0 I}{5^{3/2}R} \quad \dots(1)$$

Given $N = 50$, $I = 10A$, $R = 0.5$ m and

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\therefore B = 8.94 \times 10^{-4} \text{ T} \quad \dots(2)$$

Emf generated between the centre and the rim of the disc is

$$\begin{aligned} \xi &= \pi r^2 B f = \pi \times (0.1)^2 \times 8.94 \times 10^{-4} \times 16.66 \\ &= 468 \times 10^{-6} \text{ V} = 468 \mu\text{V} \\ &\simeq 0.5 \text{ mV} \end{aligned}$$

4. Ans (B)

$$v = (\beta/4\mu)(d^2/4 - r^2)$$

$$dv/dr = (\beta/4\mu)(-2r) = -2\beta r/4\mu$$

$$\tau = \mu(dv/dr) = \mu(-2\beta r/4\mu) = -2\beta r/4$$

For the wall, $r = d/2$

$$\tau_{\text{wall}} = \frac{-2\beta(d/2)}{4} = -\frac{\beta d}{4}$$

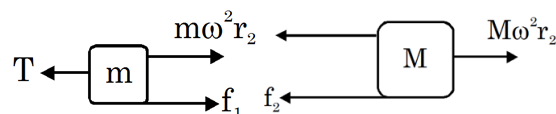
$$D_{\text{rag}} = (\tau_{\text{wall}})(\text{area}) = (\tau_{\text{wall}})(\pi dL)$$

$$(\beta d/4)(\pi dL) = \beta d^2 \pi L/4$$

5. Ans (B)

In the situation of limiting equilibrium, we have

$$T = m\omega^2 r_1 + f_1 \text{ and } T + f_2 = M\omega^2 r_2$$



[where f_1 and f_2 are frictional forces for the two blocks and the surface]

$$f_1 = \mu_1 mg \text{ and } f_2 = \mu_2 Mg$$

$\therefore$ The above two equations get reduced to

$$T = m\omega^2 r_1 + \mu_1 mg \quad \dots(i)$$

$$\text{and } T + \mu_2 Mg = M\omega^2 r_2 \quad \dots(ii)$$

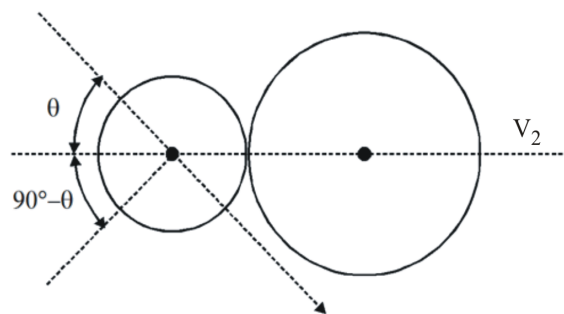
Subtracting eq. (i) from eq. (ii)

$$\mu_2 Mg = M\omega^2 r_2 - m\omega^2 r_1 - \mu_1 mg$$

$$\Rightarrow \omega^2 = \frac{g[\mu_1 m + \mu_2 M]}{Mr_2 - mr_1}$$

$$\Rightarrow \omega = \left[\frac{g[\mu_1 m + \mu_2 M]}{Mr_2 - mr_1} \right]^{\frac{1}{2}}$$

6. Ans (A)



Along line of impact

$$v_2 + v_1 \sin \theta = (u \cos \theta)$$

Since the velocity perpendicular to the line of impact remains unchanged

$$\therefore u \sin \theta = v_1 \cos \theta \quad \dots(i)$$

Finally, applying principle of conservation of linear momentum along the line of impact, we have

$$mu \cos \theta = Mv_2 - mv_1 \sin \theta \quad \dots(ii)$$

Eliminating v_1 and v_2 from Eq. (i) and (ii) we get the desired condition.

Eqs. (i) and (ii) can be solved for v_2 as follows.

$$\text{From eq. (i), } v_1 = \frac{u \sin \theta}{\cos \theta}$$

Putting the expression for v_1 in eq. (ii) we get

$$mu \cos \theta = Mv_2 - \frac{mu \sin^2 \theta}{\cos \theta} \Rightarrow v_2 = \frac{mu}{M \cos \theta}$$

Using these expressions for v_1 and v_2 in Eq. (ii),

we get

$$\frac{mu}{M \cos \theta} + \frac{u \sin^2 \theta}{\cos \theta} = u \cos \theta$$

$$\Rightarrow \frac{m + M \sin^2 \theta}{M \cos \theta} = \cos \theta$$

$$\Rightarrow m + M \sin^2 \theta = M \cos^2 \theta$$

$$\Rightarrow \sin^2 \theta (m + M) = \cos^2 \theta (M - m)$$

$$\Rightarrow \tan^2 \theta = \frac{M - m}{m + M} = \frac{1}{3} \quad \tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A,B,C,D)

$$\text{Cobalt : } E_K = 13.6 (Z - \sigma)^2$$

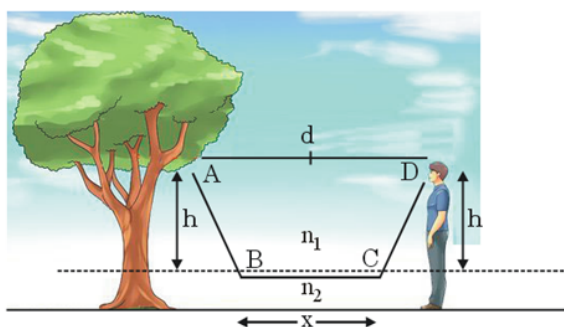
$$= 13.6(27 - 1)^2 = 9,193.6 \text{ eV}$$

$$\lambda_K = \frac{1,241}{9,193.6} \text{ nm} = 0.135 \text{ nm} = 1.35 \text{ \AA}$$

$$\text{Iron : } E_K = 13.6(26 - 1)^2 = 8,500 \text{ eV}$$

$$\lambda_K = \frac{1,241}{8,500} = 0.146 \text{ nm} = 1.46 \text{ \AA}$$

8. Ans (A,B,D)



(a) Optical path length (O.P.L.)

$$\begin{aligned} \sum_{i=1}^2 n_i s_i &= n_1 AB + n_2 BC + n_1 CD \\ &= n_1 \sqrt{(d-x)^2/4 + h^2} + n_2 x + n_1 \sqrt{(d-x)^2/4 + h^2} \end{aligned}$$

$$\text{O. P. L.} = n_2 x + n_1 \sqrt{(d-x)^2 + 4h^2} \dots (1)$$

$$(b) \frac{d(\text{O. P. L.})}{dx} = 0 \text{ (Fermat's principle)}$$

$$\text{Using (1), } n_2 - \frac{n_1(d-x)}{\sqrt{(d-x)^2 + 4h^2}} = 0$$

substituting $n_1 = 1.00020$ and $h = 2 \text{ m}$ and solving

$$d - x = 282.8 \dots (2)$$

$$\text{or } x = 500 - 282.8 = 217.2 \text{ m}$$

(c) From (2) it is observed that as d decreases, x also decreases. The smallest value for x is zero, in which case the distance of the observer from the tree would be $d = 282.8 \text{ m}$ when the mirage disappears.

9. Ans (A,B,D)

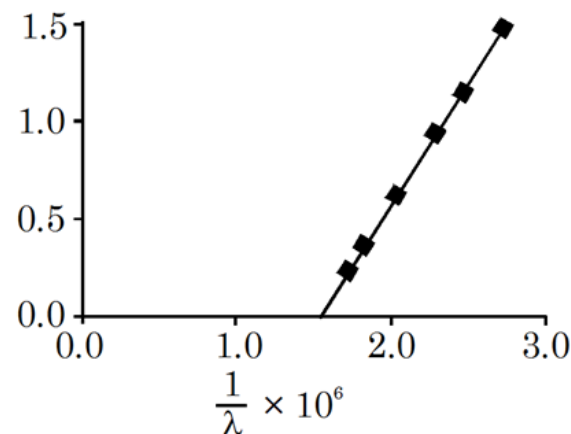
$$eV = hc/\lambda - W$$

A plot of V against $1/\lambda$ must be a straight line. The slope of the line gives hc/e , hence h can be determined.

The intercept multiplied by hc give W , the work function. The threshold frequency is given by

$$\nu_0 = \frac{W}{h}$$

V	$\lambda \times 10^{-10} \text{ m}$	$(1/\lambda) \times 10^6$
1.48	3,660	2,732
1.15	4,050	2,469
0.93	4,360	2,293
0.62	4,920	2,032
0.36	5,460	1,831
0.24	5,790	1,727



$$\text{Slope} = 1.24/10^6 = hc/e$$

$$h = 1.24 \times 10^{-6} \times e/c$$

$$= (1.24 \times 10^{-6} / (3 \times 10^8)) = 4.13 \times 10^{-15} \text{ eV-s}$$

$$\text{intercept} = 1.5 \times 10^{-6} \text{ m}^{-1}$$

$$W = hc \times \text{intercept} = 6.6 \times 10^{-34} \times 3 \times 10^8 \times 1.5 \times 10^6$$

$$= 3 \times 10^{-19} \text{ J} = 3 \times 10^{-19} / 1.6 \times 10^{-19} \text{ eV} = 1.9 \text{ eV}$$

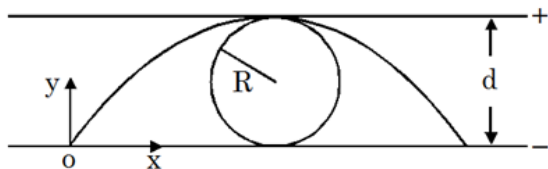
$$h\nu_0 = W$$

$$\nu_0 = W/h = 3 \times 10^{-19} / 6.6 \times 10^{-34} = 4.6 \times 10^{14}$$

$$\lambda_0 = 6.6 \times 10^{-7} \text{ m}$$

11. Ans (A,B,C,D)

Choose the origin at O. The electric field E acts along the Y -direction perpendicular to the plates which are located in the x -direction. If the component of initial velocity in the direction of B is zero then the path of electron will be contained entirely in the xy -plane.



From Lorentz force $\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$, in the component form

$$\vec{v} = v_x \hat{i} + v_y \hat{j}$$

$$F_y = m \frac{dv_y}{dt} = qE - qBv_x \quad \dots(1)$$

$$F_x = m \frac{dv_x}{dt} = qBv_y \quad \dots(2)$$

Writing for convenience

$$\omega = \frac{qB}{m} \text{ and } \gamma = \frac{E}{B} \quad \dots(3)$$

Equations (1) and (2) can be rewritten as

$$\frac{dv_y}{dt} = \gamma\omega - \omega v_x \quad \dots(4)$$

$$\frac{dv_x}{dt} = \omega v_y \quad \dots(5)$$

Differentiating (4) and using (5)

$$\frac{d^2 v_y}{dt^2} = -\omega \frac{dv_x}{dt} = -\omega^2 v_y$$

$$\text{or } \frac{d^2 v_y}{dt^2} + \omega^2 v_y = 0 \quad \dots(6)$$

With the initial conditions, $v_x = v_y = 0$, at $t = 0$,

(6) has the solution

$$v_y = A \sin \omega t \quad \dots(7)$$

where $A = \text{constant}$:

$$\therefore \frac{dv_y}{dt} = A\omega \cos \omega t = \omega\gamma - \omega v_x$$

At $t = 0$ $v_x = 0$

$$\therefore A\omega = \omega\gamma \rightarrow A = \gamma$$

$$\therefore v_y = \gamma \sin \omega t \quad \dots(8)$$

Substituting (8) into (5), integrating and using the initial condition $v_x = 0$ at $t = 0$

$$v_x = \gamma (1 - \cos \omega t) \quad \dots(9)$$

The coordinates x and y at any time t can be found out by integrating separately (8) and (9) with the initial condition $x = y = 0$ at $t = 0$

$$y = \frac{\gamma}{\omega} (\omega t - \sin \omega t) \quad \dots(10)$$

$$x = \gamma \left(t - \frac{\sin \omega t}{\omega} \right) \quad \dots(11)$$

Using (9) and (10) we get

$$v_x = \omega y \quad \dots(12)$$

The energy of the particle is unaffected in the static magnetic field. Under the electric field in the y -direction the energy picked up will be

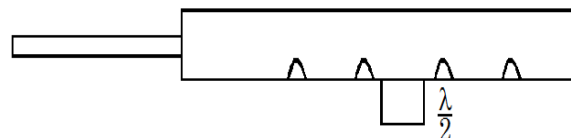
$$\frac{qVy}{d} = \frac{1}{2} mv^2 = \frac{1}{2} m (v_x^2 + v_y^2)$$

$$\text{or } v_y^2 = \frac{2qVy}{md} - \omega^2 y^2 \quad \dots(13)$$

where we have used (12).

12. Ans (A,C,D)

(a) The frequency of the rod is fixed, and so also for the air and the gas. The distance between successive heaps of cork dust is equal to the distance between two neighbouring nodes which is $\frac{1}{2}\lambda$, Figure



$$v_{\text{air}} = f\lambda_{\text{air}}$$

$$v_{\text{gas}} = f\lambda_{\text{gas}}$$

$$\therefore v_{\text{gas}} = v_{\text{air}} \frac{\lambda_{\text{gas}}}{\lambda_{\text{air}}} = 340 \times \frac{10}{8} = 425 \text{ m/s}$$

(b)

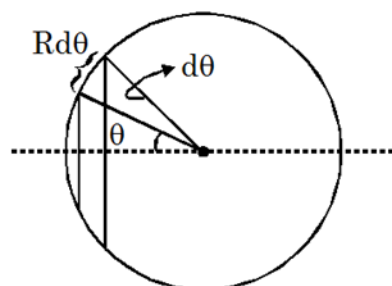
$$v_{\text{rod}} = f\lambda_{\text{rod}} = \frac{v_{\text{air}}}{\lambda_{\text{air}}} \cdot \lambda_{\text{rod}} = \left(\frac{340}{2 \times 0.08} \right) (2L_{\text{rod}})$$

$$= \left(\frac{340}{0.16} \right) (2 \times 1.2) = 5100 \text{ m/s}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (8.00)**



$$dq = \frac{q}{2\pi R} \cdot Rd\theta = \frac{q}{2\pi} \cdot d\theta$$

$$di = \frac{dq}{T} = \frac{qd\theta\omega}{2\pi 2\pi}; di = \frac{q\omega}{4\pi^2} \cdot d\theta$$

The magnetic field at O due to elementary ring is

$$dB = \frac{\mu_0 di (R \sin \theta)^2}{2R^2}$$

$$\int dB = \int_0^\pi \frac{\mu_0 \sin^2 \theta}{2R} \left(\frac{q\omega}{4\pi^2} \right) d\theta$$

$$B = \frac{\mu_0 q\omega}{16\pi R}$$

The flux of the magnetic field through the small ring

$$\phi = B\pi a^2$$

$$\phi = \pi a^2 \cdot \frac{\mu_0 q\omega}{16\pi R}; \phi = \frac{\mu_0 q\omega a^2}{16R}$$

$$|\epsilon| = \left| \frac{d\phi}{dt} \right|; |\epsilon| = \frac{\mu_0 qa^2}{16R} \alpha = 8 \text{ volt}$$

$$i = \frac{8}{1} = 8A$$

2. **Ans (5.00)**

$$\theta = 20s = \left(\frac{20}{3600} \right) \times \frac{1}{57.3} \text{ rad}$$

$$= 9.695 \times 10^{-5} \text{ rad}$$

$$\beta = \frac{\lambda}{2\mu\theta} = \frac{5.82 \times 10^{-5} \text{ cm}}{2 \times 1.5 \times 9.695 \times 10^{-5}} = 0.2 \text{ cm}$$

Number of fringes per centimetre

$$= \frac{1}{\beta} = \frac{1}{0.2} = 5$$

3. **Ans (1.35 to 1.43)**

The flow of heat in a material placed between the walls of a coaxial cylinder is given by

$$\frac{dQ}{dt} = \frac{2\pi LK}{\ln\left(\frac{r_2}{r_1}\right)} (T_1 - T_2) \dots (1)$$

Number of decays of radon atoms per second

$$dN/dt = 100 \times 10^{-3} \times 3.7 \times 10^{10}$$

$$= 3.7 \times 10^9 \text{ disintegration/second}$$

Energy deposited by α 's

$$= 3.7 \times 10^9 \times 5.5 \text{ MeV/s}$$

$$= 2.035 \times 10^{10} \text{ MeV/s} = 3.256 \times 10^{-7} \text{ J}$$

$$= 0.775 \times 10^{-3} \text{ Cal/s}$$

Using the values, $k = 0.025 \text{ Cal cm s}^{-1} \text{ C}^{-1}$,

$$L = 5 \text{ cm}, r_1 = 2 \text{ mm}$$

and $r_2 = 8 \text{ mm}$ in (1), and solving for $(T_1 - T_2)$

$$\text{we find } (T_1 - T_2) = 1.368 \times 10^{-3} \text{ } ^\circ\text{C}$$

4. **Ans (0.99)**

$$\frac{L^3}{T} = \{61.9\} (L)^{2.63} \left(\frac{M}{L^2 T^2} \right)^{0.54}$$

$$\{61.9\} = L^{1.45} T^{0.08} M^{-0.54}$$

$$\text{Ans. } 0.99$$

5. **Ans (14.72)**

Let F_C = force exerted on smaller piston at C;

$$F_C = (460) \left(\frac{240}{120} \right) = 920N$$

$$\frac{F_C}{A_C} = \frac{F_D}{A_D},$$

$$(920) / \left(\pi \left(\frac{60}{1000} \right)^2 / 4 \right) = F_D / \left[\pi \left(\frac{240}{1000} \right)^2 / 4 \right],$$

$$F_D = 14.720N.$$

6. **Ans (8.00)**

Let the distance AB = s and let the particles meet at the middle point C

where AC = BC = s/2 after time t

$$\frac{s}{2} = u_1 t + \frac{1}{2} f_1 t^2 = u_2 t + \frac{1}{2} f_2 t^2 \dots (1)$$

$$\therefore (u_1 - u_2)t = -\frac{1}{2}t^2 (f_1 - f_2)$$

$$t = -2 \frac{(u_1 - u_2)}{f_1 - f_2} \dots (2)$$

Further we are given that velocities V after travelling the distance s are same. Hence we have the following equations :

$$\therefore V^2 = u_1^2 + 2f_1 s = u_2^2 + 2f_2 s$$

$$\therefore u_1^2 - u_2^2 = -2s(f_1 - f_2)$$

$$\text{or } s = -\frac{u_1^2 - u_2^2}{2(f_1 - f_2)} \dots (3)$$

Substituting the values of s and t in (1), we get

$$\frac{s}{2} = t \left(u_1 + \frac{1}{2} f_1 t \right)$$

or

$$-\frac{(u_1^2 - u_2^2)}{4(f_1 - f_2)} = -2 \frac{(u_1 - u_2)}{(f_1 - f_2)} \left\{ u_1 - \frac{1}{2} f_1 \frac{2(u_1 - u_2)}{(f_1 - f_2)} \right\}$$

$$\therefore (u_1 + u_2) = 8 \left\{ \frac{f_1 u_1 - f_2 u_1}{(f_1 - f_2)} \right\}$$

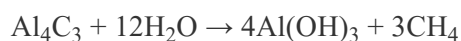
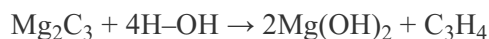
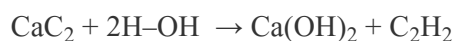
$$\text{or } (u_1 + u_2)(f_1 + f_2) = 8(f_1 u_2 - f_2 u_1)$$

$$\therefore k = 8$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A)**



2. **Ans (C)**

To stop osmosis we need to apply pressure of $(C_2 - C_1)RT$ on right side

$$\therefore (A) P_1 = 0, P_2 = (C_2 - C_1)RT \text{ No osmosis}$$

$$(B) P_1 = C_1 RT, P_2 = C_2 RT \text{ No osmosis}$$

$$(D) P_1 = 2C_1 RT, P_2 = (C_1 + C_2)RT \text{ No osmosis}$$

3. **Ans (C)**

$$A^2 = \alpha t + \beta$$

On diff. the equation we get

$$-2A dA = \alpha dt$$

$$\frac{-dA}{dt} = \frac{\alpha}{2A}$$

$$\text{rate} = \frac{\alpha}{2} [A]^{-1}$$

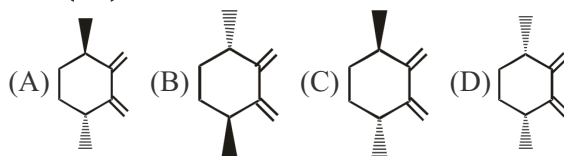
$$\therefore \text{order} = -1$$

4. **Ans (B)**

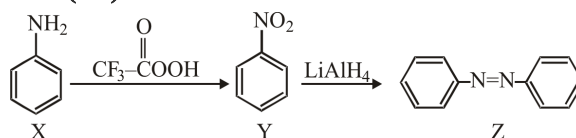
In adiabatic expansion the temp. decreases and KE ↓

$$\text{and } \Delta S_{\text{surr}} = \frac{-q}{T} = 0$$

5. **Ans (D)**



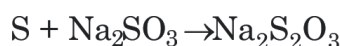
6. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-I (ii)

8. **Ans (A,B,C,D)**



9. **Ans (A,B,D)**

$$(A) W_T = W_{1 \rightarrow 2} + W_{2 \rightarrow 3} + W_{3 \rightarrow 1}$$

$$= 0 + (-P \Delta V) + \left(-2.303 nRT \log \frac{V_2}{V_1} \right) = 184.22 \text{ cal}$$

process is cyclic $\Delta U = 0$, $q = -184.22 \text{ cal}$

$$(B) W_T = W_{1 \rightarrow 2} + W_{2 \rightarrow 3} + W_{3 \rightarrow 1}$$

$$\Rightarrow (-1 \times 22.4) + [0] + \left(-2.303 \times RT \log \frac{22.4}{44.8} \right)$$

$$= -166.9 \text{ cal}$$

$$q = +166.9 \text{ cal}$$

$$(C) W_T = \frac{1}{2} \times (20 - 10) (3 - 1) \text{ L-atm} = -241.19 \text{ cal}$$

$$q = 241.19 \text{ cal}$$

$$(D) = W_{1 \rightarrow 2} + W_{2 \rightarrow 3} + W_{3 \rightarrow 4} + W_{4 \rightarrow 1}$$

$$= -5.6 \text{ L-atm} = -138.18 \text{ cal}$$

$$q = +138.18 \text{ cal}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.00)**

$\text{BaCrO}_4 \rightarrow \text{Yellow}$

$\text{Hg}_2\text{CrO}_4 \rightarrow \text{Red}$

$\text{ZnS} \rightarrow \text{White \& produces } \text{H}_2\text{S gas.}$

$\text{BaSO}_4 \rightarrow \text{White}$

$\text{BaS}_2\text{O}_3 \rightarrow \text{White and colourless gas } \text{SO}_2 \text{ comes out.}$

$\text{CH}_3\text{COOAg} \rightarrow \text{White, } \text{CH}_3\text{COOH} \uparrow$

$\text{AgNO}_2 \rightarrow \text{White but brown } \text{NO}_2 \text{ gas comes out}$

2. **Ans (4.00)**

$$Y - Y_1 = \frac{Y_2 - Y_1}{X_2 - X_1} (X - X_1)$$

$$P - 16 = \frac{4 - 16}{15 - 6} (V - 6)$$

$$3P + 4V = 72$$

$$T_{\max} = \frac{(PV)_{\max}}{nR}, T_{\max} = \frac{12 \times 9}{3 \times \frac{1}{12}}$$

$$\text{for } (PV)_{\max}, T_{\max} = 432 = 108Y$$

$$3P = \frac{72}{2}; 4V = \frac{72}{9}, \therefore Y = 4$$

$$P = 12, V = 9$$

3. **Ans (7.00)**

$$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^3$$

$$Z = 33, S = 3 + 1 = 4, n = 15 \text{ and } P = 4$$

$$\frac{33 + 4 + 15 + 4}{8} = 7$$

4. **Ans (4.00)**

$\text{O}_3, \text{H}_2\text{O}_2, \text{Cl}_2$ and Cl_2O bleaches litmus solution and converts $\text{I}^- \rightarrow \text{I}_2$.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$xf(x) - \int_0^x f(t)dt = x + \ln(\sqrt{1+x^2} - x)$$

Diff. w.r.to x

$$xf'(x) + f(x) - f(x)$$

$$= 1 + \frac{1}{(\sqrt{1+x^2} - x)} \cdot \left(\frac{x}{\sqrt{1+x^2}} - 1 \right)$$

$$\Rightarrow g(x) = 1 - \frac{1}{\sqrt{1+x^2}}$$

$$\text{Now } \because x^2 \geq 0$$

$$\therefore \sqrt{1+x^2} \geq 1$$

$$\Rightarrow g(x) \in [0, 1)$$

2. **Ans (B)**

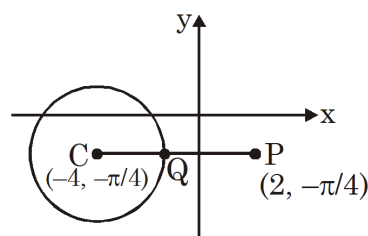
For define 1st curve

$$-1 \leq 1 - x \leq 1 \text{ and } -1 \leq 3 - x \leq 1$$

$$x \in [0, 2] \text{ and } x \in [2, 4]$$

$$\Rightarrow \text{domain of first curve is } x = 2$$

$$\Rightarrow 1^{\text{st}} \text{ curve is point } (2, \frac{-\pi}{4})$$



$$|PQ| = |PC| - r = 6 - 1 = 5$$

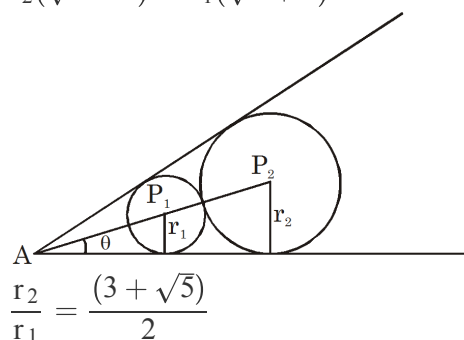
3. **Ans (D)**

$$\sin \theta = \frac{1}{\sqrt{5}} = \frac{r_1}{AP_1} = \frac{r_2}{AP_2}$$

$$AP_2 = AP_1 + r_1 + r_2$$

$$r_2\sqrt{5} = r_1\sqrt{5} + r_1 + r_2$$

$$r_2(\sqrt{5} - 1) = r_1(\sqrt{5} + 1)$$



$$\frac{r_2}{r_1} = \frac{(3 + \sqrt{5})}{2}$$

4. Ans (B)

$$M(x) = \int \frac{e^x(e^{2x} - 2)}{e^{4x} + 8e^{2x} + 4} dx$$

$$\text{Put } e^x = t$$

$$M(x) = \int \frac{t^2 - 2}{t^4 + 8t^2 + 4} dt$$

$$M(x) = \frac{1}{2} \tan^{-1} \left(\frac{e^x + 2e^{-x}}{2} \right) + c$$

$$M(0) = \frac{1}{2} \tan^{-1} \left(\frac{3}{2} \right)$$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = e^x + 2e^{-x} \Rightarrow f(\log_e 2) = 3$$

5. Ans (C)

$$\text{Equation of tangent is } y = 6x \pm \sqrt{36a^2 + b^2}$$

This is normal which pass through $(-2, 0)$

$$0 = -12 \pm \sqrt{36a^2 + b^2}$$

$$36a^2 + b^2 = 144$$

Now AM $\geq$ GM

$$\frac{36a^2 + b^2}{2} \geq \sqrt{36a^2 b^2}$$

$$72 \geq 6ab$$

$$ab \leq 12$$

6. Ans (B)

$$(1-x)t^2 + (2+x^2)t - (x^2 + 2x) = 0$$

$t = a$ and b distinct roots

$$a + b = \frac{2+x^2}{x-1} \text{ and } a \cdot b = \frac{x^2+2x}{x-1}$$

$$\Rightarrow a + b - ab = -2$$

$$ab - a - b = 2$$

$$(a-1)(b-1) = 3$$

$$a = 4, b = 2 \text{ or } a = 2, b = 4$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B,C,D)

(A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then z_1 and z_2 subtend right angle at circumcentre origin

$\therefore$ the chord joining z_1 and z_2 will subtend an angle θ at 'z' such that

$$\begin{cases} \theta = \pi/4 & \text{if } |z| = 1 \\ \theta < \pi/4 & \text{if } |z| > 1 \\ \theta > \pi/4 & \text{if } |z| < 1 \end{cases}$$

$$(B) |z_1 z_2 + z_2 z_3 + z_3 z_1| = |z_1| \cdot |z_2| \cdot |z_3|$$

$$\left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right|$$

$$(C) \text{ Let } z = \left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 \cdot z_2 \cdot z_3} \right)$$

$$\text{then } \bar{z} = z$$

(D) The triangle formed by joining z_1, z_3 and z_2 is isosceles and right angled at z_3 .

8. Ans (B,C)

$$2\cos^2 2x - 2\cos 2x \cos\left(\frac{2014\pi^2}{x}\right) = -2\sin^2 2x$$

$$\Rightarrow \cos 2x \cos \frac{2014\pi^2}{x} = 1$$

$\cos 2x, \cos\left(\frac{2014\pi^2}{x}\right)$ should both be simultaneously 1 or -1.

Positive solutions : $\pi, 19\pi, 53\pi, 1007\pi$

$$\text{sum} = 1080\pi$$

9. Ans (B,C)

$$30! = 2^{26} \times 3^{14} \times 5^7 \times 7^4 \times 11^2 \times 13^2 \times 17^1 \times 19^1 \times 23^1 \times 29^1$$

Contains 10 primes and there is 1-1 correspondence between picking a pair of $(m, n) = 1$ and $mn = 30!$

and forming a subset from these 10 primes. Half of these pair has $m > n$ and the other half $m < n$. So the

$$\text{answer is } \frac{2^{10}}{2} = 512.$$

10. **Ans (A,B,D)**

(A) for $\lambda \neq -6$ lines are non parallel, hence unique solution for all α and β .

(B) If $\alpha + 4\beta = 0$ and $\lambda = -6$ lines are coincident, hence infinite solutions.

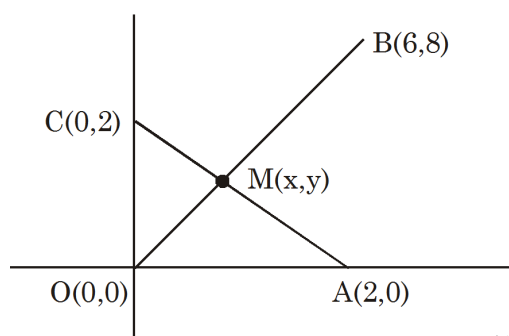
(D) if $\alpha + 4\beta \neq 0$ and $\lambda = -6$ lines are parallel, hence no solutions.

11. **Ans (A,C,D)**

Let $O(0, 0)$, $A = (2, 0)$, $B = (6, 8)$, $C = (0, 2)$

Now for $f(x, y)$ to be minimum, (x, y) must be the point of intersection of OB and AC .

$$\therefore f(x, y)_{\min} = OB + AC = 10 + 2\sqrt{2}$$



Which occurs at $x = \frac{6}{7}$, $y = \frac{8}{7}$

12. **Ans (B,C,D)**

$$t_1 t_2 = 2, t_1 + t_2 + t_3 = 0, R(3t_3^2, 6t_3)$$

Coordinates of point T are $(at_1 t_2, a(t_1 + t_2)) \equiv (2a, -at_3) \equiv (6, -3t_3)$

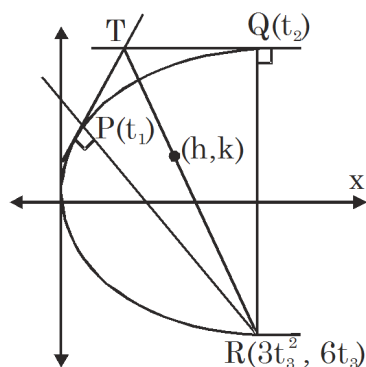
Let midpoint of TR be (h, k)

$$\Rightarrow 2h = 3t_3^2 + 6 \quad \dots (i)$$

$$2k = 3t_3 \quad \dots (ii)$$

$$\therefore 2h = \frac{4k^2}{3} + 6$$

$$k^2 = \frac{3}{2}(h - 3)$$



$$y^2 = \frac{3}{2}(x - 3) \text{ (Parabola)}$$

Now interpret.

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3.00)**

$$I_n = \int_0^{\frac{3\pi}{2}} (\ln |\sin x|) \cos(2nx) dx$$

applying integration by parts

$$I_n = \left\{ \ln |\sin x| \cdot \frac{\sin 2nx}{2n} \right\}_0^{\frac{3\pi}{2}} - \int_0^{\frac{3\pi}{2}} \frac{\cot x \cdot \sin 2nx}{2n} dx$$

$$I_n = 0 - \frac{1}{2n} I_n'$$

$$I_n' = \int_0^{\frac{3\pi}{2}} \frac{\cos x \cdot \sin 2nx}{\sin x} dx$$

$$I_n' - I_{n-1}' = \int_0^{\frac{3\pi}{2}} \frac{\cos x (\sin 2nx - \sin(2n-2)x)}{\sin x} dx$$

$$= \int_0^{\frac{3\pi}{2}} \frac{2 \cos x \cdot \cos(2n-1)x \sin x}{\sin x} dx$$

$$I_n' - I_{n-1}' = \int_0^{\frac{3\pi}{2}} 2 \cos(2n-1)x \cdot \cos x dx = 0$$

$$I_n' = I_{n-1}' = I_{n-2}' = \dots = I_1'$$

$$I_n' = \int_0^{\frac{3\pi}{2}} \frac{\sin 2x \cos x}{\sin x} dx$$

$$= \int_0^{\frac{3\pi}{2}} 2 \cos^2 x dx = \int_0^{\frac{3\pi}{2}} (1 + \cos 2x) dx$$

$$= \frac{3\pi}{2}$$

$$I_n = -\frac{3\pi}{4n}$$

$$12I_3 = -3\pi$$

$$16I_2 = -6\pi$$

$$\Rightarrow 12I_3 - 16I_2 = 3\pi$$

2. Ans (1.00)

$$f''(x) - 5f'(x) + 6f(x) = 0$$

$$\Rightarrow f''(x) - 2f'(x) = 3(f'(x) - 2f(x))$$

$$\frac{2f'(x) - f''(x)}{2f(x) - f'(x)} = 3$$

$$\ln(2f(x) - f'(x)) = 3x + c$$

$$2f(x) - f'(x) = k \cdot e^{3x}$$

$$\because f(0) = 1, f'(0) = 0 \Rightarrow k = 2$$

$$\Rightarrow f'(x) - 2f(x) = -2e^{3x}$$

$$e^{-2x}f(x) = -2 \int e^{3x} \cdot e^{-2x} dx$$

$$= -2e^x + c$$

$$\because f(0) = 1 \Rightarrow c = 3$$

$$f(x) = 3e^{2x} - 2e^{3x}$$

$$f'(x) = 6(e^{2x} - e^{3x})$$

$$\begin{array}{c} \uparrow \quad \downarrow \\ \hline x = 0 \end{array}$$

$$\Rightarrow \text{maximum of } f(x) \text{ is } f(0) = 1$$

3. Ans (50.00)

$$\text{Let } g(x) = (x-4)p(x) - 1$$

$$\because g(5) = g(6) = g(7) = \dots = g(105) = 0$$

$$\therefore g(x) = A \cdot (x-5)(x-6)(x-7) \dots (x-105)$$

$$\because g(4) = -1$$

$$\therefore A = \frac{1}{(101)!}$$

$$\Rightarrow (x-4)p(x) - 1$$

$$= \frac{1}{(101)!} (x-5)(x-6)(x-7) \dots (x-105)$$

$$\text{Now put } x = 106$$

$$(102)p(106) - 1 = 1 \Rightarrow p(106) = \frac{1}{51}$$

$$m = 1; n = 51$$

4. Ans (0.00)

$$\vec{c} = \frac{\vec{a} - \mu \vec{b}}{4} \text{ and } \vec{a} \cdot \vec{b} = 0$$

$$\text{Now, } (\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$$

$$\Rightarrow (\vec{b} - \vec{a}) \cdot \left(\vec{b} + \frac{\vec{a} - \mu \vec{b}}{4} \right) = 0$$

$$\Rightarrow (4 - \mu)b^2 = a^2 \quad (\because \mu < 4) \quad \dots(i)$$

$$\text{Again } 4 \left| \vec{b} + \vec{c} \right|^2 = \left| \vec{b} - \vec{a} \right|^2$$

$$\Rightarrow 4 \left| \frac{(4 - \mu)\vec{b} + \vec{a}}{4} \right|^2 = \left| \vec{b} - \vec{a} \right|^2$$

$$\Rightarrow 4 \left(\frac{4 - \mu}{4} \right)^2 b^2 + \frac{a^2}{4} = b^2 + a^2$$

$$\Rightarrow ((4 - \mu)^2 - 4)b^2 = 3a^2 \quad \dots(ii)$$

$$(i) \text{ and } (ii) \text{ we get } \frac{(4 - \mu)^2 - 4}{4 - \mu} = \frac{3}{1}$$

$$\Rightarrow \mu^2 - 5\mu = 0$$

$$\Rightarrow \mu = 0 \text{ or } 5 \text{ but as } \mu < 4, \text{ so } \mu = 0.$$

5. Ans (1.00)

$$q = \sum_{r=1}^{\infty} \frac{1}{2^r} \left(1 - \frac{1}{2^r} \right)^{2021}$$

$$q = \sum_{r=1}^{\infty} \frac{1}{2^r} \left[\sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \left(\frac{1}{2^r} \right)^k \right]$$

$$q = \sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \sum_{r=1}^{\infty} \left(\frac{1}{2^r} \cdot \frac{1}{2^{rk}} \right)$$

$$q = \sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \left[\frac{1}{2^{k+1}} + \frac{1}{2^{2(k+1)}} + \frac{1}{2^{3(k+1)}} + \dots \right]$$

$$q = \sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \left(\frac{\frac{1}{2^{k+1}}}{1 - \frac{1}{2^{k+1}}} \right)$$

$$q = \sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \left(\frac{1}{2^{k+1} - 1} \right)$$

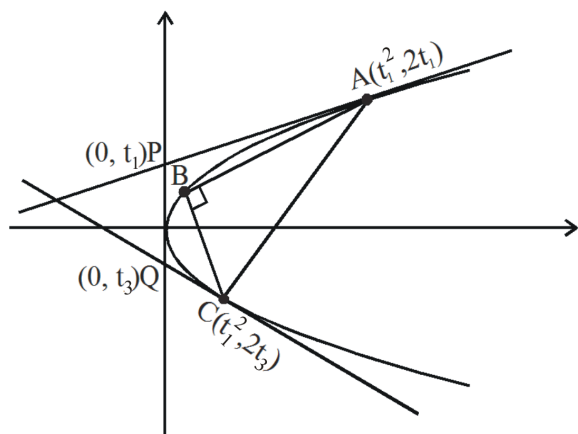
$$q = \sum_{r=0}^{2021} (-1)^r \cdot \frac{{}^{2021}C_r}{(2^{r+1} - 1)} \quad \dots(1)$$

$$q = 1 - p$$

$$(p + q) = 1$$

6. Ans (4.00)

Let $A(t_1^2, 2t_1)$; $B(t_2^2, 2t_2)$; $C(t_3^2, 2t_3)$



$\therefore AB \perp BC$ and length of intercept = PQ

$\therefore (\text{slope of } AB) \cdot (\text{slope of } BC) = -1$

$$\frac{2}{(t_1 + t_2)} \cdot \frac{2}{(t_2 + t_3)} = -1 \text{ and } PQ = |t_1 - t_3|$$

$$\Rightarrow t_2^2 + (t_1 + t_3)t_2 + (t_1 t_3 + 4) = 0$$

$\therefore t_2 \in \mathbb{R}$

$\therefore D \geq 0$

$$(t_1 + t_3)^2 - 4(t_1 t_3 + 4) \geq 0$$

$$(t_1 - t_3)^2 \geq 16$$

$$|t_1 - t_3| \geq 4$$

$$\Rightarrow \text{minimum length of intercept} = |t_1 - t_3| = 4$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,C	A,C	A,B,C	A,B,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.80	2.00	2.00	1.90 to 1.92	3.00	15.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	2	2	4	5	7	5

PART-2 : CHEMISTRY

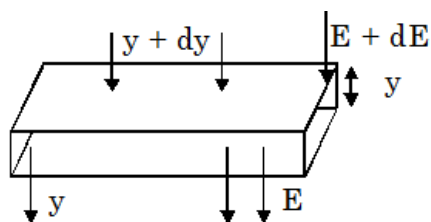
SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C	D	A,C	A,B,C	A,D	B
SECTION-II	Q.	1	2	3	4	5	6
	A.	3.33	80.20	0.70 to 0.72	3.10 to 3.14	7.75	11.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	7	7	5	7	1	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,C,D	A,B	A,B	A,B,D	A,B,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	20.00	0.25	3.14	0.53	40.00	108.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	1	8	3	1	1	4

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (B,C)

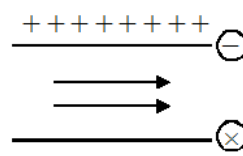

 $\rho = 4y$, ρ increases as y increases

 $\Rightarrow$ electric field is along -ve y -direction

$$(-dE)A = \frac{\rho dy A}{\epsilon_0} \Rightarrow \frac{-dE}{dy} = \frac{4y}{\epsilon_0}$$

$$E = -\frac{2y^2}{\epsilon_0}$$

3. Ans (A,C)

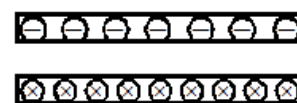
 Current per unit length = σV


$$B = \left(\frac{\mu_0 K}{2} \right) 2 = \mu_0 \sigma V$$

$$B = \frac{\mu_0 \sigma V}{2}$$

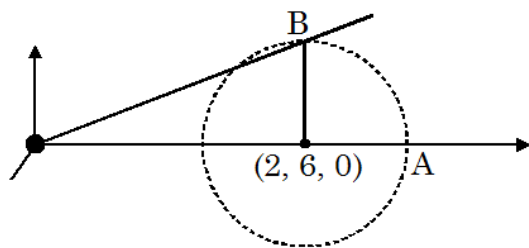
Outward current

Inward current



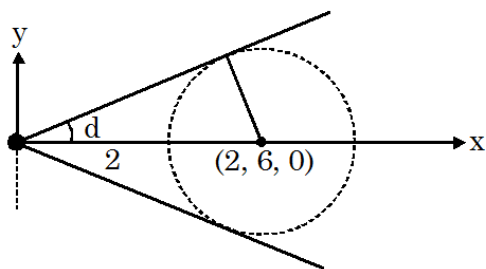
$$\begin{aligned} \text{For per unit area } \vec{L} \times \vec{B} &= \sigma V_1 \times \frac{\mu_0 \sigma V}{2} \\ &= \frac{\mu_0 \sigma^2 V^2}{2} (\text{Upward}) \end{aligned}$$

4. Ans (A,B,C)



$$B \tan \theta = \frac{1}{2}$$

$$\oint \vec{B} \cdot d\vec{l} = \frac{\mu_0 I}{2\pi} \tan^{-1} \left(\frac{1}{2} \right)$$



$$\phi = \frac{\pi}{6}$$

$$\left[\oint \vec{B} \cdot d\vec{l} \right]_{\max} = \frac{\mu_0 I}{6}$$

6. Ans (A,B,C,D)

$$E \propto \frac{z^2}{n^2} \quad E_{pr} \propto \frac{1}{n}$$

$$p \propto v \propto \frac{z}{n} \quad \frac{p}{E} \propto n$$

$$r \propto \frac{n^2}{z} \quad E_r \propto z \text{ (independent of } n)$$

$$pr \propto n$$

PART-1 : PHYSICS

SECTION-II

1. Ans (1.80)

E_0 = Electric field deep inside in each set

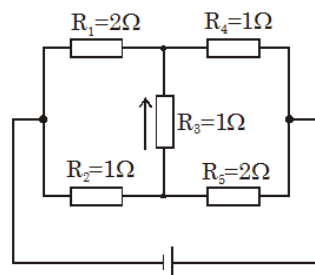
$$E_1 = \frac{E_0}{360} \times 60 = \frac{E_0}{6} \text{ (Equilateral triangle)}$$

$$E_2 = \frac{E_0}{360} \times 100 = \frac{3E_0}{10}$$

$$\text{Ratio} = \frac{3E_0}{10} \times \frac{6}{E_0} = 1.80$$

2. Ans (2.00)

The electric current can go through the diagram by three ways: through the upper branch (resistors R_1 and R_4), through the middle branch (resistors R_2 , R_3 and R_4) and through the bottom branch (resistors R_2 and R_5). The voltage drop on each resistor is equal to the product of its resistance and the current strength going through it and the sum of voltage drops on all the paths has to be equal to the voltage on the current source. Thus, $I_1 R_1 + I_4 R_4 = I_2 R_2 + I_3 R_3 + I_4 R_4 = I_2 R_2 + I_5 R_5 = U_0$. From these equations we can derive the expressions $I_1 R_1 = I_2 R_2 + I_3 R_3$ and $I_3 R_3 + I_4 R_4 = I_5 R_5$, which describe voltage drops on the left and right side of the diagram. In addition we can write down equations for current strengths: $I_1 = I_0 - I_2$, $I_4 = I_1 + I_3$ and $I_5 = I_2 - I_3$.



Let us notice that if $R_1 = R_2$ and $R_4 = R_5$ then by symmetry considerations $R_3 = 0$ should apply.

It is clear that $R_3 = 1\Omega$. Let us first suppose that $R_1 = R_4 = R_3$. In this case the voltage drop in the upper branch is expressed as $(2I_1 + I_3)R_3 = U_0$ from which follows that $I_1 = 6$ A and $I_2 = 4$ A. Observing the path of the current through the middle branch we can write down that $I_2 R_2 + U_3 + (I_1 + I_3)R_3 = U_0$ from which follows that $R_2 = R_5 = 1\Omega$. Such diagram satisfies the symmetry mentioned before, therefore the assumption made is not true. Let us now suppose that $R_2 = R_5 = R_3$. For the bottom branch of the diagram we can write $(2I_2 - I_3)R_3 = U_0$ meaning $I_2 = 8$ A and $I_1 = 2$ A. Voltage drops on the left side of the diagram must satisfy the equation $I_1 R_1 = (I_2 + I_3)R_3$ from which follows that $R_1 = 5\Omega$. In this case, however, the voltage drop on the upper branch of the scheme is $(2I_1 + I_3)R_1 = 30$ V $\neq U_0$. Let us now suppose that $R_1 = R_5 = R_3$. The voltage drop on the middle branch of the scheme is $I_2 R_2 + U_3 + (I_0 - I_2 + I_3)R_2 = U_3 + (I_0 + I_3)R_2 = U_0$ meaning $R_2 = R_4 = 1\Omega$. Now we are dealing with a situation already looked at. The only possibility left is that $R_2 = R_4 = R_3$. From the left side of the diagram we get that $(I_0 - I_2)R_1 = I_2 R_3 + U_3$ and from the right side $(I_2 - I_3)R_1 = U_3 + (I_0 - I_2 + I_3)R_3$. Based on these $(I_0 - I_3)R_1 = 2U_3 + (I_0 + I_3)R_3$ meaning $R_1 = 2\Omega$. It is easy to make sure that the values found satisfy all the equations for voltage drops.

3. Ans (2.00)



$$\sin i = \mu_1 \sin r_1 \quad \dots\dots(i)$$

$$\mu_1 \cos r_1 = \mu_2 \sin r_3 \quad \dots\dots(ii)$$

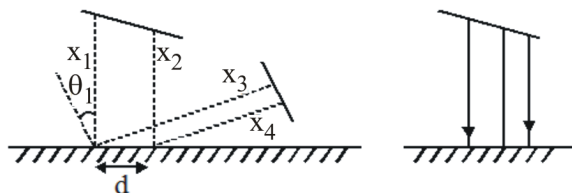
$$\mu_2 \cos r_3 = \mu_2 \sin r_5 \quad \dots\dots(iii)$$

$$\mu_3 \cos r_5 = \cos i \quad \dots\dots(iv)$$

$$\mu_2^2 + \sin^2 i + \cos^2 i = \mu_1^2 + \mu_3^2$$

$$\mu_2 = \sqrt{\mu_1^2 + \mu_3^2 - 1} = \sqrt{5 - 1} = 2$$

4. Ans (1.90 to 1.92)



$$x_3 - x_1 = n\lambda$$

$$x_4 - x_2 = (n - 1)\lambda$$

$$(x_4 - x_3) + (x_1 - x_2) = -\lambda$$

$$-d \sin \theta_2 + d \sin \theta_1 = -\lambda$$

$$d = \frac{-\lambda}{(\sin \theta_1 - \sin \theta_2)}$$

$$d = \frac{\lambda}{(\sin \theta_2 - \sin \theta_1)} = \frac{700}{\frac{\sqrt{3}}{2} - \frac{1}{2}} = \frac{1400}{(\sqrt{3} - 1)} \times 10^{-9} \text{ m}$$

$$= \frac{1.4}{(\sqrt{3} - 1)} \mu\text{m} = \frac{1.4}{0.732} = 1.91$$

5. Ans (3.00)

$$\rho_{\text{rel}} = \frac{A}{A - W}$$

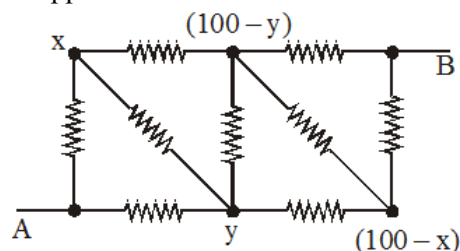
$$\frac{1}{\rho_{\text{rel}}} = 1 - \frac{W}{A} \left[\rho_{\text{rel}} = \frac{10}{5} = 2 \right]$$

$$-\frac{\rho_{\text{rel}}}{\rho_{\text{rel}}^2} \Delta \rho_{\text{rel}} = \frac{\Delta W}{W} + \frac{\Delta A}{A}$$

$$\left(\frac{\Delta \rho_{\text{rel}}}{\rho_{\text{rel}}} \right) = \frac{\Delta W}{W} + \frac{\Delta A}{A} = 1\% + 2\% = 3\%$$

6. Ans (15.00)

Lets assume an external potential of $V = 100$ Volt is applied across A and B.



Applying junction rule

$$\frac{x - 100}{R} + \frac{x - y}{R} + \frac{(x + y - 100)}{R} = 0 \Rightarrow x = \frac{200}{3}$$

$$\frac{y - 100}{R} + \frac{(y - x)}{R} + \frac{(2y - 100)}{R} + \frac{(y + x - 100)}{R} = 0$$

$$\Rightarrow x = 60V$$

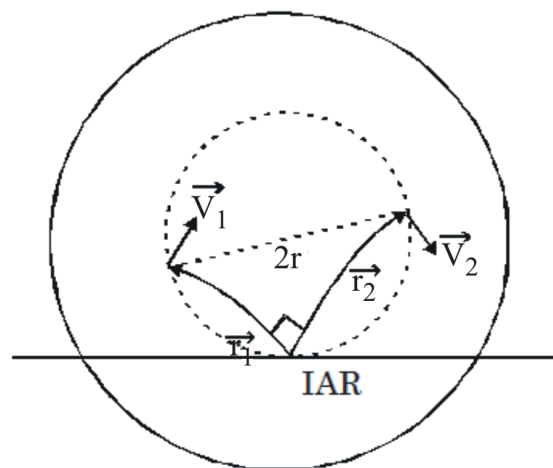
$$I = \frac{100 - x}{R} + \frac{100 - y}{R} = \frac{220}{R}$$

$$V = IR_{\text{eq}} \Rightarrow 100 = \frac{220}{3R} R_{\text{eq}} \Rightarrow R_{\text{eq}} = \frac{15R}{11}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (2)



$$\left\{ (2r)^2 = r_1^2 + r_2^2 \right\} \times w$$

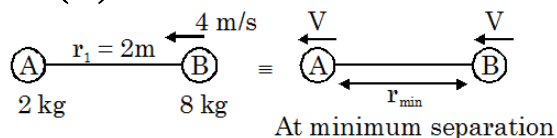
$$(2r w)^2 = (r_1 w)^2 + (r_2 w)^2$$

$$(2V)^2 = V_1^2 + V_2^2$$

$$V = \frac{\sqrt{V_1^2 + V_2^2}}{2}$$

$$= \frac{\sqrt{7+9}}{2} = \frac{4}{2} = 2 \text{ m/s}$$

2. Ans (2)



Applying conservation of momentum

$$9 \times 4 = 10V \Rightarrow V = \frac{16}{5} \text{ m/s}$$

Introducing potential energy function

$$\frac{-dU}{dr} = 8 \Rightarrow U = -8r$$

Applying conservation of mechanical energy,

$$\frac{1}{2} \times 8 \times (4)^2 - 8 \times 2 = \frac{1}{2} \times 10 \times \left(\frac{16}{5}\right)^2 - 8r_{\min}$$

$$\Rightarrow r_{\min} = 0.4$$

$$5 \times r_m = 2.0$$

3. Ans (4)

After a long time disc C will come to rest and only A and B will be moving in opposite directions so from conservation of momentum A and B will have same speed Now from conservation of energy we get.

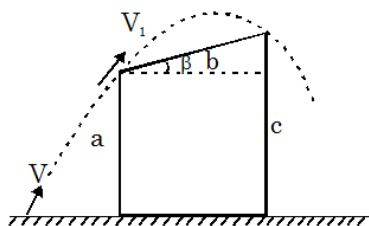
$$mgh = \left(\frac{1}{2}MV^2\right) \times 2$$

$$V = \sqrt{\frac{mgh}{M}}$$

$$= \sqrt{\frac{4 \times 10 \times 1}{2000 \times 2}} = \frac{1}{10} \text{ m/s}$$

$$40V = 4.00$$

4. Ans (5)



$$\frac{V_1^2}{g(1 - \sin \beta)} = b(\text{Maximum range})$$

$$\frac{V_1^2}{g\left(1 + \frac{(c-a)}{b}\right)} = b$$

$$\Rightarrow V_1^2 = g(b + c - a)$$

$$V^2 = V_1^2 + 2ag$$

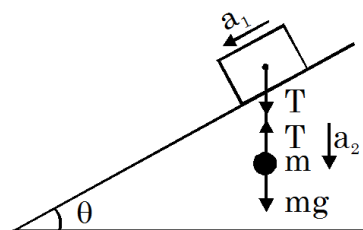
$$V^2 = g(b + c - a) + 2ag = g(a + b + c)$$

$$V = \sqrt{g(a + b + c)}$$

$$= \sqrt{10 \times 10} = 10 \text{ m/s}$$

$$\frac{V}{2} = \frac{10}{2} = 5 \text{ m/s}$$

5. Ans (7)



$$Mg \sin \theta + T \sin \theta = Ma_1 \quad \dots\dots(i)$$

$$mg - T = ma_2 \quad \dots\dots(ii)$$

$$\text{Also } a_1 \sin \theta = a_2 \quad \dots\dots(iii)$$

From (i) and (ii)

$$(M - m)g \sin \theta = Ma_1 + ma_2 \sin \theta$$

$$(M + m)g \sin \theta = Ma_1 + ma_1 \sin^2 \theta$$

$$\Rightarrow a_1 = \frac{(M + m)g \sin \theta}{(M + m \sin^2 \theta)}$$

$$a_2 = \frac{(M + m)g \sin^2 \theta}{(M + m \sin^2 \theta)}$$

$$= \frac{(12 + 8) \times 10 \times \frac{1}{4}}{(12 + 8 \times \frac{1}{4})}$$

$$= \frac{50}{14} = \frac{25}{7} \text{ m/s}^2$$

6. Ans (5)

Pressure of gas remains constant during process

$$P = P_0, \Delta Q = nC_p \Delta T = n\left(\frac{1}{2} + 1\right)P \Delta T = \left(\frac{3}{2} + 1\right)nR \Delta T$$

$$H = \frac{5}{2}P \Delta V = \frac{5}{2}P_0 \Delta V = \frac{5}{2} \times 10 \times 0.4 \times 10^{-3} = 100 \text{ J}$$

PART-2 : CHEMISTRY

SECTION-I

3. Ans (A,C)

No transfers one e^- in non-dative made to Fe.

In $[\text{Fe}(\text{NH}_3)_6]^{+2}$, electrons remain in both t_{2g} & e_g .

Ni^{+4} has d^6 configuration hence after e^- pairing, it's diamagnetic.

In $[\text{Pt} \text{Cl}_3(\text{C}_2\text{H}_4)]^-$, one Pt-Cl bond is weaker [trans to C_2H_4 due to synergic bonding]

4. **Ans (A,B,C)**

SiF_4 undergoes partial hydrolysis where 3 moles are consumed forming 2 moles of Fluorosilicic acid & a mole of silicic acid.



5. **Ans (A,D)**

Zn is more volatile than Ag.

Anode mud contains these impurities in unoxidised form.

6. **Ans (B)**

According to Langmuir's theory, reactivity of a site is independent of the status of surrounding sites.

During desorption, the time upto which a molecule can remain adsorbed is called average

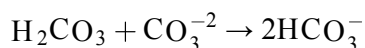
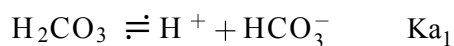
$$\text{time } \left[t_{\text{avg}} = \frac{1}{k} \right]$$

ZSM-5 is shape selective catalyst.

PART-2 : CHEMISTRY

SECTION-II

3. **Ans (0.70 to 0.72)**



$$\frac{K_{a1}}{K_{a2}} = \frac{4 \times 10^{-8}}{4 \times 10^{-12}} = 10^4$$

$$t = 0.1 \quad \text{C} \quad \text{O}$$

$$t = t.0.1 - \text{C} \quad \text{O} \quad 2\text{C}$$

$$6.4 = 7.4 + \log \frac{2\text{C}}{0.1 - \text{C}} \Rightarrow \frac{1}{10} = \frac{2\text{C}}{0.1 - \text{C}}$$

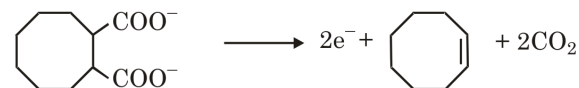
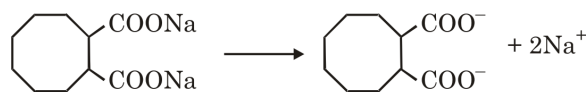
$$20\text{C} = 0.1 - \text{C}$$

$$\text{C} = \frac{0.1}{21} = \frac{x}{150} \times 1000$$

$$x = 0.714 \text{ m.mole}$$

4. **Ans (3.10 to 3.14)**

$$x = 3$$



$$\therefore 1 \text{ mole requires } \rightarrow 2\text{F}$$

$$\therefore \frac{2.44}{244} \text{ requires } \rightarrow '0.02\text{F}'$$

$$y \times \frac{15}{100} = 0.02 \Rightarrow y = 0.133$$

5. **Ans (7.75)**

$$x = 5 \quad z = 2.75$$

$$y = 3 \quad w = 3$$

6. **Ans (11.00)**

Total geometrical isomers of reactant = 4

product formula of $[\text{M}(\text{gly})_2\text{bc}]$

Total geometrical isomers of product = 7

$$\therefore \text{total G.I.} = 11$$

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (7)**

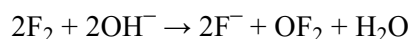
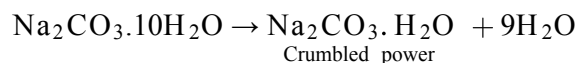
$$1, 2, 3, 4, 5, 8, 9$$

2. **Ans (7)**

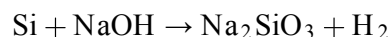
$$1, 2, 3, 4, 5, 6, 9$$

3. **Ans (5)**

True : (ii), (iii), (iv), (v), (viii)



HgO is yellow ppt.



Thermal stability of KOH is higher as compared to NaOH

In Castner-Kellner's cell, hydrogen has high overpotential on Hg due to which Na is deposited on Hg cathode.

4. **Ans (7)**

$$12\sqrt{3}a^2 + 6a^2 = \text{total surface area}$$

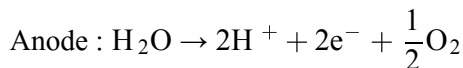
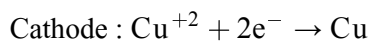
$$\therefore \text{Answer} = 6.75 \text{ cm}^2$$

5. **Ans (1)**

$$8.64 = \frac{108}{96500} \times 10 \times 965 \times \phi$$

$$\phi = 0.8$$

In CuSO_4 :



$\therefore$ 2F produce 2 mole H^+

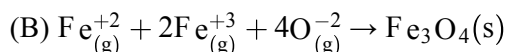
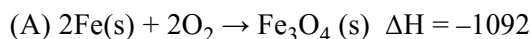
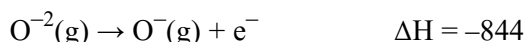
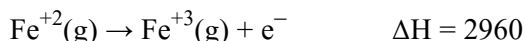
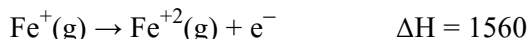
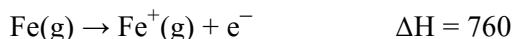
$$\therefore \frac{0.965 \times 1250}{96500} \times 0.8 \text{ produce} = 0.01 \text{ mole } \text{H}^+$$

$$\therefore \text{Molarity of } \text{H}^+ = \frac{0.01}{0.1} = 0.1\text{M}$$

$$\therefore \text{pH} = 1$$

$$\text{Ans.} = 1$$

6. **Ans (6)**



$$\Delta H = ?$$

Equation (A) – (B) operation is obtained by

previous seven equations.

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,C,D)**

$$\text{Volume} = [\vec{b} \times \vec{c}, \vec{c} \times \vec{a}, \vec{a} \times \vec{b}] = [\vec{a} \vec{b} \vec{c}]^2$$

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 2 + \sin \theta & \cos \theta & \sin 2\theta \\ \sin(\theta + \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) & \sin(2\theta + \frac{4\pi}{3}) \\ \sin(\theta - \frac{2\pi}{3}) & \cos(\theta - \frac{2\pi}{3}) & \sin(2\theta - \frac{4\pi}{3}) \end{vmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 + R_3 \\ \begin{vmatrix} 2 + \sin \theta & \cos \theta & \sin 2\theta \\ -\sin \theta & -\cos \theta & -\sin 2\theta \\ \sin(\theta - \frac{2\pi}{3}) & \cos(\theta - \frac{2\pi}{3}) & \sin(2\theta - \frac{4\pi}{3}) \end{vmatrix} \end{aligned}$$

$$\begin{aligned} R_1 &\rightarrow R_1 + R_2 \\ \begin{vmatrix} 2 & 0 & 0 \\ -\sin \theta & -\cos \theta & -\sin 2\theta \\ \sin(\theta - \frac{2\pi}{3}) & \cos(\theta - \frac{2\pi}{3}) & \sin(2\theta - \frac{4\pi}{3}) \end{vmatrix} \end{aligned}$$

$$\begin{aligned} &= 2 \sin 2\theta \cos\left(\theta - \frac{2\pi}{3}\right) - 2 \cos \theta \cdot \sin\left(2\theta - \frac{4\pi}{3}\right) \\ &= 2 \sin 2\theta \cos\left(\theta - \frac{2\pi}{3}\right) - 2 \cos \theta \cdot \sin\left(2\theta + \frac{2\pi}{3}\right) \\ &= \sin\left(3\theta - \frac{2\pi}{3}\right) + \sin\left(\theta + \frac{2\pi}{3}\right) - \sin\left(3\theta + \frac{2\pi}{3}\right) - \sin\left(\theta + \frac{2\pi}{3}\right) \\ &= 2 \cos 3\theta \cdot \sin\left(-\frac{2\pi}{3}\right) = -\sqrt{3} \cos 3\theta \end{aligned}$$

$$\text{Value} = \frac{1}{6} (3 \cos^2 2\theta) = \frac{1}{2} \cos^2 3\theta$$

$$\therefore f(\theta) = \frac{1}{2} \cos^2 3\theta$$

$$(A) \frac{1}{2} \cos^2 3\theta = \frac{1}{4} \quad \cos 3\theta = \pm \frac{1}{\sqrt{2}}$$

$$3\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$$

$$\theta = \frac{\pi}{12}, \frac{\pi}{4}, \frac{5\pi}{12}$$

$$(B) \text{Max. value of } f(\theta) = \frac{1}{2}$$

$$(C) \sum_{n=1}^5 f\left(\frac{n\pi}{33}\right) = \frac{1}{2} \sum_{n=1}^5 \cos^2\left(\frac{n\pi}{11}\right)$$

$$= \frac{1}{4} \sum_{n=1}^5 \left(1 + \cos \frac{2n\pi}{11}\right)$$

$$= \frac{5}{4} + \frac{1}{4} \sum_{n=1}^5 \cos \frac{2n\pi}{11} = \frac{5}{4} - \frac{1}{8} = \frac{9}{8}$$

$$\sqrt{f(\theta)} = \frac{1}{\sqrt{2}} |\cos 3\theta|$$

$$\text{Non-diff. at } \theta = \frac{\pi}{6}$$

2. Ans (A,B)

$$\frac{dy}{dx} = \frac{2 \left(\frac{x^2+y^2}{x^2-y^2} \right) \pm \sqrt{4 \left(\frac{x^2+y^2}{x^2-y^2} \right)^2 - 4}}{2}$$

$$\frac{dy}{dx} = \frac{x+y}{x-y} \text{ or } \frac{x-y}{x+y}$$

$$\frac{dy}{dx} = \frac{x+y}{x-y} \Rightarrow xdy - ydy = xdx + ydx$$

$$\frac{xdx + ydy}{x^2 + y^2} = \frac{xdy - ydx}{x^2 + y^2}$$

$$\therefore \ln(x^2 + y^2) = 2 \tan^{-1} \frac{y}{x} + C$$

Case-II : $\frac{dy}{dx} = \frac{x-y}{x+y}$

$$xdy + ydx = xdx - ydy$$

$$2xy = x^2 - y^2 + C$$

3. Ans (A,B)

PQ is focal chord of parabola.

Circle with diameter PQ always touches the line $x + 5 = 0$.

$$\therefore x + 5 = 0, \text{ Directrix of parabola}$$

Equation of axis is $y = \lambda$

$$\therefore \text{Focus is } (1 - \lambda, \lambda)$$

$$\text{Now } \frac{|1 - \lambda + 5|}{1} = \frac{8}{2} \therefore \lambda = 10, 2$$

$$\therefore \text{Focus} = (-9, 10) \text{ and } (-1, 2)$$

Equation of parabola

$$(y - 10)^2 = -8(x + 7) \text{ and } (y - 2)^2 = 8(x + 3)$$

4. Ans (A,B,D)

$$AB - BP = 2I, B = \text{adj}(B)$$

$$\text{Let } B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\therefore b = c = 0$$

$$\text{adj}(B) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad a = d$$

$$\therefore B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$$

$$\therefore aA - aP = 2I$$

$$aA = aP + 2I$$

$$\text{Let } P = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

$$\text{where } \alpha\delta - \beta\gamma = 3$$

$$\alpha + \delta = 4$$

$$aA = \begin{bmatrix} a\alpha + 2 & a\beta \\ a\gamma & a\delta + 2 \end{bmatrix}$$

$$\therefore a^2 |A| = a^2 (\alpha\delta - \beta\gamma) + 2a(\alpha + \delta) + 4$$

$$a^2 |A| = 3a^2 + 8a + 4$$

$$\text{If } |A| = 15 \therefore 12a^2 - 8a - 4 = 0$$

$$a = -1/3, 1$$

$$\therefore \text{Tr}(B) = 2a = -2/3, 2$$

$$\text{If } \text{Tr}(B) = 4$$

$$2a = 4$$

$$a = 2$$

$$\therefore 4|A| = 12 + 16 + 4$$

$$\therefore |A| = 8 \therefore \text{adj}(\text{adj}(A)) = 8$$

5. Ans (A,B,C,D)

$$f(x) = x^3 + 5x + 1$$

$$g'(y) = \frac{1}{f'(x)} = \frac{1}{3x^2 + 5}$$

when $y = 1$ then $x = 0$

$$\therefore g'(1) = \frac{1}{5}$$

$$h(g(x)) = x$$

$$\therefore x \rightarrow f(x)$$

$$h(g(x)) = f(x)$$

$$\therefore h(x) = f(f(x))$$

$$h(0) = f(f(0)) = 7$$

$$h(g(2))$$

$$f(f(g(2))) = f(2) = 8 + 10 + 1 = 19$$

$$h'(x) = f'(f(x)) \cdot f'(x)$$

$$h'(1) = f'(f(1)) \cdot f'(1)$$

$$= f'(7) \cdot f'(1)$$

$$= (147 + 5)(3 + 5) = 152 \times 8 = 1216$$

6. Ans (A,B,C,D)

$$X = {}^{20}C_3 = 1140$$

$$Y = \frac{{}^{20}C_1 \cdot {}^{16}C_2}{3} = 800$$

$$Z = {}^{20}C_3 - 10 \times {}^{18}C_1 = 1140 - 180 = 960$$

$$Y \cap Z = Y - 10 \times 14$$

$$= 800 - 140$$

$$= 660$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (20.00)

$$100n a_n = 100(n-2)a_{n-1} + 1, a_0 = 0, a_1 = \frac{1}{100}$$

Multiply both side by $(n-1)$

$$100n(n-1)a_n = 100(n-1)(n-2)a_{n-1} + n-1$$

$$\text{Let } 100n(n-1)a_n = b_n$$

$$\therefore 100(n-1)(n-2) = b_{n-1}, b_0 = 0$$

$$\therefore b_n = b_{n-1} + n-1$$

$$b_n - b_{n-1} = n-1 \quad \forall n \geq 1$$

$$b_1 - b_0 = 0$$

$$b_2 - b_1 = 1$$

$$b_n - b_{n-1} = n-1$$

$$\therefore b_n - b_0 = \frac{n(n-1)}{2}$$

$$\therefore b_n = \frac{n(n-1)}{2}$$

$$\therefore 100n(n-1)a_n = \frac{n(n-1)}{2} = b_n$$

$$\therefore a_n = \frac{1}{200} \quad \forall n \geq 2$$

$$\left[\sum_{n=0}^{4044} a_n \right] = \left[\frac{4044}{200} \right] = 20$$

2. Ans (0.25)

$$f(x, y) = axy + by + cx + d$$

$$f(x, y) = x(ay + c) + by + d$$

$$\text{Observe } f(x+2, y) - f(x+1, y) = f(x+1, y) - f(x, y)$$

$$\text{i.e. } f(x+2, y) = 2f(x+1, y) - f(x, y)$$

$$\text{Similarly } f(x, y+2) = 2f(x, y+1) - f(x, y)$$

$$\therefore f(5, 5) = 2f(5, 4) - f(5, 3)$$

$$= 2[2f(4, 4) - f(3, 4)] - [2f(4, 3) - f(3, 3)]$$

$$= 4f(4, 4) - 2f(3, 4) - 2f(4, 3) + f(3, 3)$$

$$= \frac{4 \times 9}{4 \times 4} - \frac{2 \times 9}{3 \times 4} - \frac{18}{4 \times 3} + \frac{9}{3 \times 3}$$

$$= \frac{9}{36} = \frac{1}{4} = 0.25$$

3. **Ans (3.14)**

$$I = \int_1^3 \left(\tan^{-1} \left(\frac{(x-2)^2 - (x-2)}{(x-2)^3 + 1} \right) + \tan^{-1} \left(\frac{1}{(x-2)^2} \right) \right) dx$$

Let $x - 2 = t$

$$I = \int_{-1}^1 \left(\tan^{-1} \frac{t(t-1)}{1+t^3} + \tan^{-1} \frac{1}{t^2} \right) dt$$

$$I = \int_{-1}^1 \left(\tan^{-1} t^2 - \tan^{-1} t + \tan^{-1} \frac{1}{t^2} \right) dt$$

$$I = \frac{\pi}{2} \times 2 = \pi = 3.14$$

4. **Ans (0.53)**

$$n(s) = \frac{6!}{2!2!2! \cdot 3!} \times 3! = 90$$

Now 90 = All three gets correct pair

+ Exactly one of the person gets correct pair

+ No. one gets correct pair

$$\therefore 3! + (3!) \times 6 + x = 90$$

$$x = 48$$

$$\therefore \text{required prob.} = \frac{48}{90} = \frac{8}{15} = 0.53$$

5. **Ans (40.00)**

$$1 + x + x^2 + \dots + x^{2n} = \frac{x^{2n+1} - 1}{x - 1}$$

$$1 - x + x^2 - \dots + x^{2n} = \frac{x^{2n+1} + 1}{x + 1}$$

$$\therefore \text{Product} = \frac{(x^{2n+1})^2 - 1}{x^2 - 1} = \frac{(x^2)^{2n+1} - 1}{x^2 - 1}$$

$$\frac{(x^2 - 1)(1 + x^2 + (x^2)^2 + \dots + (x^2)^{2n})}{x^2 - 1}$$

$$= 1 + (x^2) + (x^2)^2 + \dots + (x^2)^{2n}$$

$$\therefore \text{Sum of coeff. } 2n + 1 = 81$$

$$\therefore n = 40$$

6. **Ans (108.00)**

$$l = 2r \sin \left(\frac{\theta}{2} \right)$$

$$p = 2r \sin 18^\circ$$

$$q = 2r \sin 30^\circ$$

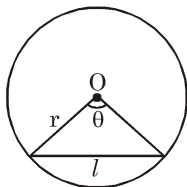
$$p + q = 2r (\sin 18^\circ + \sin 30^\circ)$$

$$= 2r \left(\frac{\sqrt{5}-1}{4} + \frac{1}{2} \right)$$

$$p + q = 2r \left(\frac{\sqrt{5}+1}{4} \right)$$

$$= 2r \sin 54^\circ$$

$\therefore p + q$ subtends an angle 108° at centre.



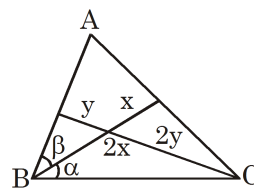
PART-3 : MATHEMATICS

SECTION-III

1. **Ans (1)**

$$\tan \beta = \frac{y}{2x}$$

$$\tan \alpha = \frac{2y}{2x}$$



$$\tan (B) = \tan (\alpha + \beta) = \frac{\frac{y}{2x} + \frac{y}{x}}{1 - \frac{y^2}{2x^2}} = \frac{3xy}{2x^2 - y^2}$$

$$\therefore \cot B = \frac{2x^2 - y^2}{3xy}$$

$$\text{Similarly } \cot C = \frac{2y^2 - x^2}{3xy}$$

$$\cot B + \cot C = \frac{x^2 + y^2}{3xy} \geq \frac{2}{3}$$

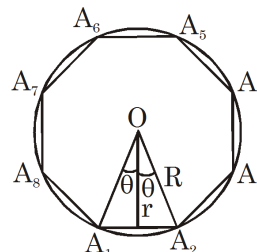
2. **Ans (8)**

$$x + iy = \frac{K_r + 2i}{K_r^2 + 4}$$

$$\therefore x = \frac{K_r}{K_r^2 + 4}, \quad y = \frac{2}{K_r^2 + 4}$$

$$\therefore x^2 + y^2 = \frac{y}{2}$$

$\therefore$ all vertices lies on a circle whose radius is $1/4$.



$$\angle A_1 O A_2 = \frac{2\pi}{8} = \frac{\pi}{4}$$

$$\therefore \theta = \frac{\pi}{8}$$

$$\cos \frac{\pi}{8} = \frac{r}{R}$$

$$\therefore r = R \cos \frac{\pi}{8} = \frac{1}{4} \cos \frac{\pi}{8}$$

$$\therefore \text{Area of circle} = \pi r^2$$

$$= \pi \cdot \frac{1}{16} \cos^2 \frac{\pi}{8} = \frac{\pi}{16} \left(\frac{1 + \cos \frac{\pi}{4}}{2} \right)$$

$$= \frac{\pi}{32} \left(1 + \frac{1}{\sqrt{2}} \right) = \frac{\pi(\sqrt{2} + 1)}{32\sqrt{2}}$$

$$\therefore a = 2$$

$$b = 32$$

$$\frac{b}{2a} = \frac{32}{4} = 8$$

3. Ans (3)

$$f(x) > g(x)$$

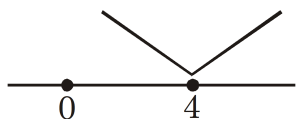
$$|x| + \frac{4}{|x|} > p|x| + 2 \quad \forall x \in \mathbb{R} - \{0\}$$

$$p < \frac{|x| + \frac{4}{|x|} - 2}{|x|}$$

$$p < 1 + \frac{4}{(x)^2} - \frac{2}{|x|} \quad \forall x \in \mathbb{R} - \{0\}$$

$$\text{Consider } h(x) = 1 + \frac{4}{x^2} - \frac{2}{x}, x > 0$$

$$h'(x) = \frac{2(x-4)}{x^3}$$



$$\text{Now, } h(4) = 1 + \frac{1}{4} - \frac{1}{2} = \frac{3}{4}$$

$$p < h(x)$$

$$\therefore 0 < p < \frac{3}{4}$$

$$a = 0, b = 3/4$$

$$\therefore a + 4b = 3$$

4. Ans (1)

$$\frac{2x}{25} + \frac{2y}{16} \cdot \frac{dy}{dx} = 0$$

$$\text{Let } A(x_1, y_1)$$

$$\therefore \text{Slope of normal at } A = \frac{25y_1}{16x_1}$$

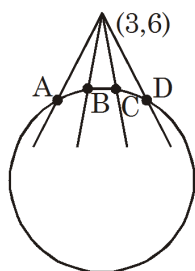
$$\therefore \frac{25y_1}{16x_1} = \frac{y_1 - 6}{x_1 - 3}$$

$$25x_1y_1 - 75y_1 = 16x_1y_1 - 96x_1$$

$$\therefore \text{Conic } 9xy + 96x - 75y = 0$$

$$\therefore (p, q) = \left(\frac{25}{3}, \frac{-32}{3} \right)$$

$$\frac{p-q}{19} = 1$$



5. Ans (1)

$$\text{Let point P on the line is } (2\lambda + 1, -\lambda - 1, \lambda)$$

The foot of perpendicular Q is given by

$$\frac{x - 2\lambda - 1}{1} = \frac{y + \lambda + 1}{1} = \frac{z - \lambda}{1} = \frac{-(2\lambda - 3)}{3}$$

$$\therefore Q \text{ Lies on } x + y + z = 3 \text{ and } x - y + z = 3$$

$$\Rightarrow x + z = 3 \text{ and } y = 0$$

$$y = 0 \Rightarrow \lambda + 1 = \frac{-2\lambda + 3}{3} \Rightarrow \lambda = 0$$

$$\Rightarrow Q \text{ is } (2, 0, 1)$$

6. Ans (4)

$$\sqrt{3}(\bar{a} \cdot \bar{b}) + |\bar{b} - (\bar{a} \cdot \bar{b})\bar{a}| = 1 + |\bar{b}|^2$$

$$\sqrt{3}(\bar{a} \cdot \bar{b}) + |\bar{a} \times (\bar{b} \times \bar{a})| = 1 + |\bar{b}|^2$$

$$\sqrt{3}(\bar{a} \cdot \bar{b}) + |\bar{a}| |\bar{b} \times \bar{a}| \sin 90^\circ = 1 + |\bar{b}|^2$$

$$\sqrt{3}(\bar{a} \cdot \bar{b}) + |\bar{a} \times \bar{b}| = 1 + |\bar{b}|^2$$

$$\sqrt{3} |\bar{b}| \cos \theta + |\bar{b}| \sin \theta = 1 + |\bar{b}|^2$$

$$\underbrace{\sqrt{3} \cos \theta + \sin \theta}_{\leq 2} = \underbrace{|\bar{b}| + \frac{1}{|\bar{b}|}}_{\geq 2}$$

$$\therefore \theta = \frac{\pi}{6} \quad \therefore \operatorname{cosec}^2 \theta = 4$$